BIOL-1 Practice Test 02

Modules 5-6: Membranes and Metabolism

Instructions: This practice test covers material from Modules 5 (Membranes) and 6 (Metabolism). Answer all questions to the best of your ability.

✂️ TEAR-OFF ANSWER SHEET **Write your name and date above. Turn in this page when instructed.**
Multiple-choice answers 1. ______ 2. ______ 3. ______ 4.
______ 5. ______ 6. ______ 7. ______ 8. ______ 9. ______ 10.
______ 11. ______ 12. ______ 13. ______ ## Fill-in answers 1.
______________________________ 2.
______________________________ 3.
______________________________ 4.
______________________________ 5.
______________________________ 6.
______________________________ 7.
______________________________ 8.
______________________________ 9.
____
One answer per line; use the question numbers printed in the booklet.

QUESTION BOOKLET — FREE RESPONSE **Answer the free-response questions on the lined space provided.** ## Part C: Short Answer *Answer each question in 2-3 complete sentences.* **23.** Explain the difference between passive transport and active transport. Give one example of each.

24. Describe what happens to a cell placed in a hypertonic solution. How would this differ for a plant cell versus an animal cell?

25. Explain how enzymes speed up chemical reactions and why they are specific to their substrates.

26. Describe the structure of ATP and explain how it stores and releases energy for the cell.

--- *End of Practice Test 02*

QUESTION BOOKLET — MULTIPLE CHOICE AND FILL-IN **Question booklet: answer the questions below using the tear-off sheet for Parts A and B.** ## Part A: Multiple Choice *Choose the best answer for each question.* ### Module 5: Membranes **1.** The cell membrane is best described by the: A) Fluid mosaic model B) Rigid barrier model C) Lock and key model D) Simple lipid model **2.** The cell membrane is primarily composed of: A) Proteins only B) Phospholipid bilayer C) Carbohydrates D) Nucleic acids **3.** Phospholipids spontaneously form a bilayer in water because: A) They are charged molecules B) They have hydrophilic heads and hydrophobic tails C) They are completely hydrophobic D) They are all the same size **4.** Cholesterol in the cell membrane functions to: A) Produce ATP B) Store genetic information C) Help maintain membrane fluidity D) Transport molecules across the membrane **5.** Which transport process requires ATP? A) Diffusion B) Facilitated diffusion C) Osmosis D) Active transport **6.** When a cell is placed in a hypotonic solution, water will: A) Move into the cell B) Not move C) Move in both directions equally D) Move out of the cell **7.** A cell in an isotonic solution will experience: A) Net water movement into the cell B) Cell lysis C) No net water movement D) Net water movement out of the cell --- ### Module 6: Metabolism **8.** Enzymes function by: A) Providing energy for reactions B) Decreasing activation energy C) Increasing activation energy D) Changing the equilibrium of reactions **9.** The molecule that stores and transfers energy in all cells is: A) Glucose B) DNA C) ATP D) NADH **10.** The part of an enzyme where the substrate binds is called the: A) Active site B) Product site C) Inhibitor site D) Allosteric site **11.** When an enzyme loses its shape due to heat or pH changes, this is called: A) Activation B) Catalysis C) Inhibition D) Denaturation **12.** ATP releases energy when: A) A phosphate bond is broken (hydrolysis) B) It binds to DNA C) It is stored in the cell D) A phosphate group is added **13.** Which statement about enzymes is TRUE? A) Enzymes are made of carbohydrates B) Enzymes increase the activation energy C) Enzymes are used up in reactions D) Enzymes can be reused --- ## Part B: Fill in the Blank *Write the correct term in the blank.* **14.** The phospholipid bilayer is described as a __________________________ mosaic model. **15.** The process by which water moves across a membrane is called __________________________. **16.** Transport that requires ATP is called __________________________ transport. **17.** A solution with a lower solute concentration than the cell is called __________________________. **18.** The specific reactant that an enzyme acts on is called the __________________________. **19.** Energy of motion is referred to as __________________________ energy. **20.** __________________________

inhibition occurs when a molecule binds to the active site and blocks the substrate. **21.** The sum of all chemical reactions in an organism is called _________________________. **22.** Reactions that release energy by breaking down complex molecules are called _________________________ reactions. ---